

1 Enhanced Split Function

Purpose

You use the split function if you would like to shorten the lead time of operations or you would like to improve the capacity utilization of several workplaces/machines available for production.

In contrast to the simple split function, the enhanced split function provides the following features:

- You can split an operation into split operations with different target quantities and setup times.
- You can split an operation that has already been split (split operation) into further split operations with different target quantities and setup times.
- You can delete individual splits.

Integration

This function splits an operation into two or more operations. An operation can be split using the following functions:

- [Order sequencing](#).
- [Edit operations](#).
- [Graphic Planning](#): can be called via the context menu of the operation.

Data is always uploaded to the PPS/ERP system according to the order/operation number of the original operation.

Requirements

An operation can only be split if the following conditions are met:

- The length of the split number is set to 1 (maximum 9 splits) or 2 (maximum 99 splits) in the [Basic settings](#).
- In the [Basic settings](#), you must set the option "Enhanced split function" to "J - Yes".
- The option "May be split" is set for the operation and the field "Max. number of splits" includes a value greater than 2.
- The operation to be split cannot be a merged operation.



You can also split operations that require batch management. This function is only available if the extension [splitmplop](#) is enabled.

Split operation

Function authorization	op.split
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You can split an operation using the function "split operation".

To perform splits, select the respective operation and start the split function using the context menu or by clicking the respective button. A dialog opens where the planner may define into how many operations the operation is to be split and what quantities are planned for each split operation.

Field description of "split operation"

The dialog shows different information depending on whether the operation is split for the first time or if splitting is performed for an operation that has already been split.

In the field descriptions below, it is specified for each field if the field is available when the OP is

- (1) split for the first time or
- (2) when the OP is further split.

MES order number (1) (2)

Operation that you want to split. In case of a first split, it is the operation that may be split (split master); for the following splits, it is the split master.

Target quantity (P) (1) (2)

Target quantity and unit of the split master according to the specifications made in the pool of orders. The target quantity and the other quantities generally refer to the primary quantity of the operation.

Actual quantity (1) (2)

(Yield) quantity that has been uploaded/confirmed so far. This quantity is entered for the split operations. The quantity is also used to update the split master status.

It includes the actual quantity of operations that have already been finished as well as the actual quantity of running or interrupted operations.

Remaining quantity for split OPs (1) (2)

For all split OPs that have not yet been finished (split OPs with control indicator S, V, L or U) with the difference resulting from "target quantity minus actual quantity" being > 0: sum total of remaining quantities.

Operations with a difference between target quantity and actual quantity that is less than 0 are not used. This difference results from overproduction. Such a surplus is no longer produced and, as a result, no longer used for further splitting.

Difference (1) (2)

This quantity results from:

"Target quantity" (P) minus "actual quantity" minus "remaining quantity of split OPs"

If this quantity is unequal 0, the text "DIFFERENCE" is shown in red. You can therefore easily identify whether/when the target quantity of the split master is entirely distributed.

Max. number of splits (1) (2)

The maximum number of splits defined for the split master.

Previous number of splits (1) (2)

Number of splits that have been performed so far. This number specifies the total number of split operations and the splits that must be performed (splits included in the list)

Remaining run time (1) (2)

Remaining run time of the split master according to the remaining run time formula.

Setup time (1) (2)

Target setup time of the split master according to the specifications made in the pool of orders. The setup time is taken over into the list during splitting and can be adjusted there individually for each split.

The information listed above generally refers to the operation to be split or to the split master, provided that the operation has already been split.

If the operation has already been split, i.e. it is now the split master, the below-mentioned information is shown for the split operation that is used for further splitting. If the operation is split for the first time, these fields are *not* displayed:

Split OP (2)

Split operation that is to be split or that is to be used for further splitting.

Target quantity (P) (2)

Target quantity of the split operation.

Actual quantity (2)

(Yield) quantity uploaded/confirmed so far for this split operation

Interactive input options of the planner

If you enter the "number of splits" and, if required, the quantity that you want to split, and if you then press the "split" button, the splits are generated in the list. You can still change the target quantity and the setup time.

Number of splits (1) (2)

Planners can now specify how many splits they want to create.

By default, the field is assigned the number of splits that are still possible (maximum number of splits minus current number of splits).

Quantity to be split (1) (2)

In addition, the planner can define a split quantity. In this case, the individual splits are assigned this quantity in the table.

If the quantity to be split is 0, the remaining quantity is distributed evenly among the splits. However, the quantity may still be changed for each split.

You can add splits until the maximum number of splits of the split master is reached.

Generate OP with difference quantity (1)

This checkbox is only visible if the OP has not yet been split. The option is therefore not set by default.

If the checkbox is enabled, another split operation is shown in the table in addition to the split OPs. Condition: the total target quantity of the split OPs to be generated (included in the list) is less than the remaining quantity of the operation to be split.

For this operation, you can change all target data, except the target quantity itself. The target quantity cannot be changed, because it is automatically calculated: remaining quantity calculated using the quantity of the operation and the total of all target quantities of the split OPs to be generated.

Set off target qty. of the split OP (2)

This checkbox is only visible if the OP is a split operation. The option is therefore not set by default.

If this checkbox is set, the target quantity of the split OP is updated, i.e. the target quantity of the split OP is reduced by the total target quantity of the split operations to be generated. However, this is only the case if this sum does not exceed the remaining quantity of the split operation (the remaining quantity results from the difference between target quantity and actual quantity). Otherwise, you cannot enable the checkbox.

Table columns

Split (1) (2)

The system generates the future split number. You cannot change the number.

The next free number is used as split number, i.e. it is checked which number is currently the highest split number.

Target quantity (P) (1) (2)

Target quantity of the split operation. The planner may change the target quantity, provided it is not the split generated due to the option "Generate OP with difference quantity".

The target quantity of the new entries results from the "quantity to be split".

Setup time (1) (2)

Target setup time of the split operation. The planner may change the setup time.

For new entries, the setup time of the split master is preassigned.

Remaining run time (1) (2)

Remaining run time (remaining processing time) according to the remaining run time formula that is defined for the split master taking into account the target quantity of the split operation. HYDRA calculates this remaining run time and it cannot be changed by the planner.

The totals row shows:

- Column "Split": number of entries,
- Column "Target quantity": Total of the target quantity,
- Column "Setup time": Total of setup times,
- Column "Remaining run time": Total of remaining run times.

An error message is output if the entered number of splits (existing and new splits) exceeds the maximum number of splits.

It is allowed to enter an (admissible) number of splits and a split quantity that is greater than the target quantity of the operation. However, a respective information saying "difference" is displayed for the user.

Editing of split entries

Existing entries can be edited or deleted using the "edit" or "delete" button. Multiple selections are supported for deletion.

Once entries have been confirmed by clicking OK, they are updated in the relevant dialog.

The target quantity cannot be changed if the checkbox "Generate OP with difference quantity" is set.

Split result

The entries are checked, once the dialog "split operation" has been confirmed by clicking OK. Error messages are shown if necessary:

- Splits having a target quantity 0 are not allowed.

- In general, the maximum number of splits is checked against the split master.

Then the system generates the split operation(s). They are directly generated in the database.

One or several new operations are generated by splitting. Their ID differs from the original split master by the split number. There are the following operations, for example:

- 123456789012 0100 = Split master
- 123456789012 0100 1 = Split operation 1
- 123456789012 0100 2 = Split operation 2

Additional information



After splitting, you can only make postings for the generated operation splits. The original operation (split master) is marked as *locked* and no postings can be made for this operation. It is no longer shown in the backlog of orders.

The HYDRA Shop Floor Scheduling module no longer shows the split master and, therefore, the master cannot be planned any longer, but only its split operations.

Changes to operation splits, in particular changes in planning (e.g. changes to planned dates or changes to the planned workplace) do not affect the split master.

Please keep in mind that after an operation has been split, any new changes from the ERP system to this original operation ("split master") will have no effect on the individual operations split off.

If the target quantity update is enabled, the operation splits will not be updated, but only the split master.

Cancel operation splits

Function authorization	op.splitrelease
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A split operation can be canceled by clicking the button , if the split operation has not yet been started.

In contrast to the simple split function, the enhanced split function only cancels, i.e. deletes the selected split operation. The split master and the other split operations remain unchanged.



Please note: If a split operation is deleted, its target quantity will not be transferred to other, still existing split operations.

If the last split operation is deleted, the split master becomes a "normal" operation.

Displaying operation splits

Whether or not split operations are displayed depends on the context of the relevant application.

Split master:

- The original operation (split master) is not shown in graphic planning.
- The split master is shown in the following applications:
 - Edit operations
 - Order information

Split operations:

- The [Graphic planning](#) generally shows operation splits. In contrast to this, the split master is no longer shown after splitting.
- The single operations split off are shown in the following applications, provided that the relevant option has been selected:
 - [Edit operations](#)
 - [Order information](#)

Posting for split operations

Where split operations are concerned, the log record is ONLY generated for the split operation; no log record is generated for the split master itself.

Only the status of the split master is updated. The following events trigger an update:

- interrupt operation
- log off operation

Uploading part quantities to the PPS/ERP system

The values recorded or posted for split operations are generally uploaded to the PPS/ERP system with reference to the split master.